Sweta Bisen

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Professional Summary

Innovative and detail-oriented **Data Scientist** with **3 years of experience** in **Machine Learning, Deep Learning, NLP, and Generative AI**. Skilled in developing and deploying scalable AI systems using **Azure and AWS**. Experienced in **LLMs, Retrieval-Augmented Generation (RAG)**, and **Agentic AI architectures** with frameworks such as **LangChain, LangFlow, CrewAI, and AutoGen**. Delivered production-grade AI solutions for **healthcare, pharma, and finance** domains, improving automation, compliance, and decision-making efficiency.

Education

Bachelor of Science(BSc) in PCM

Master of Science (M.Sc.) in Chemistry
Rashtrasant Tukadoji Maharaj Nagpur University, Maharashtra, India

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Professional Experience

Data Scientist

Bluegem IT Solution Pvt. Ltd. | Oct 2022 – Present
Duration: 3+ Years

Key Responsibilities:

- Designed and deployed **ML/DL models** for regression, classification, and clustering using **scikit-learn, TensorFlow, and PyTorch**.
- Built **Retrieval-Augmented Generation (RAG)** pipelines using **Azure OpenAI, Azure AI Search, and FAISS**.
- Developed **Agentic AI systems** using **LangChain** and **LangFlow** for autonomous reasoning and orchestration.
- Created **ETL workflows** with **PySpark** and **Apache Airflow** for automated data processing.
- Deployed applications using **Docker, Kubernetes, FastAPI**, and **Azure ML** integrated with **CI/CD pipelines**.

- Developed **Streamlit dashboards** and **Power BI reports** for visualization and stakeholder interaction.
- Automated retraining and deployment pipelines, reducing manual intervention by **40%**.

Skills & Technologies

Category	**Tools & Technologies**
Programming Languages	Python (primary), SQL
Python Frameworks	Pandas, NumPy, Polars, Scikit-learn, XGBoost, SciPy, PyTorch, TensorFlow, Hugging Face Transformers
Machine Learning	Supervised & Unsupervised Learning (Regression, Classification, Clustering), Model Training & Evaluation (Precision, Recall, F1-score, AUC-ROC, RMSE, MAE), Hyperparameter Tuning (Grid/Random Search, Optuna), Cross-Validation, Feature Engineering, A/B Testing, Statistical Validation (t-test, ANOVA)
Deep Learning	Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Transformers, BERT, Vision Transformers (ViT), Transfer Learning, Fine-tuning, Sequence-to-Sequence Modeling
Generative AI (GenAI)	Large Language Models (LLMs: GPT, Llama, Mistral, Claude), Prompt Engineering, Chain-of-Thought (CoT), Few-Shot & Zero-Shot Learning, Retrieval-Augmented Generation (RAG), Hallucination Mitigation,

	Citation Guardrails, Output Structuring (JSON, XML), LLM Fine-tuning (LoRA, QLoRA),Copilot Studio
Agentic AI	Multi-Agent Orchestration, Autonomous Task Execution, Tool Calling, Memory Management, Agent Swarms, State Management, Planning & Reasoning Agents — using LangChain, LangFlow, CrewAI, AutoGen
	Transformers Agents 2.0, ChatDev, Meta-Agent, Mistral Agent
NLP & Text Analytics	Named Entity Recognition (NER), Text Classification, Summarization, Sentiment Analysis, Semantic Similarity, Topic Modeling, Tokenization, Embeddings (Sentence-BERT, OpenAI, Bedrock), Document Intelligence
Visualization & BI	Matplotlib, Seaborn, Plotly, Power BI, Streamlit (interactive dashboards)
Cloud Platforms	Amazon Web Services (AWS), Azure
Azure AI & ML Services	Azure Machine Learning (Azure ML Studio), Azure OpenAI Service, Azure AI Search Azure Document Intelligence, Azure Functions, Azure Kubernetes Service (AKS), Azure Blob Storage, Azure Data Lake, Azure DevOps (CI/CD Pipelines), Azure Monitor, Azure AI Studio, Azure Cognitive Services

AWS AI & ML Services	Amazon SageMaker (model training, hosting, pipelines), Amazon Bedrock (LLM access: Claude, Llama, Titan, etc.), Amazon Textract (document extraction), Amazon OpenSearch Service (hybrid search: BM25 + vector), AWS Lambda, Amazon API Gateway, Amazon S3, Amazon EKS (Elastic Kubernetes Service), Amazon EC2 (GPU instances), AWS Glue (ETL), Amazon CloudWatch (monitoring)
ETL & Big Data Processing	PySpark (Spark SQL, DataFrames), Apache Airflow (workflow orchestration)
Model Deployment & MLOps	Docker, Kubernetes (EKS, AKS), FastAPI, Flask, ONNX, TensorRT, Model Registry, CI/CD (Azure DevOps, GitHub Actions), Automated Retraining, Model Monitoring, Drift Detection, Infrastructure as Code (Terraform – basic)
Vector Databases & Search	FAISS (local), Amazon OpenSearch (hybrid: keyword + vector), Azure AI Search, Chroma (conceptual), Weaviate (familiarity)
Local LLM & Inference	Ollama, LM Studio, GGUF, Llama.cpp, Hugging Face Transformers (local inference), Quantization (4-bit, 8-bit), Model Offloading (CPU/GPU)
Development Practices	Test-Driven Development (TDD) – PyTest, unittest, Code Quality – Black, Pylint, isort, Git/GitHub, GitHub Copilot (AI-assisted coding), Agile/Scrum, Code Reviews, Refactoring, Documentation
Healthcare Standards & APIs	FHIR, SMART-on-FHIR, RxNorm/RxNav, OpenFDA, DailyMed, ClinicalTrials.gov, FAERS, HL7

Retrieval-Augmented Generation (RAG)	End-to-End RAG Pipeline Design, Document Chunking (semantic, fixed-size, overlap), Embedding Models (OpenAI, Bedrock, Sentence Transformers), Vector Search (FAISS, OpenSearch, Azure AI Search), Hybrid Search (BM25 + Vector), Re-Ranking (Cohere, Cross-Encoders), Dynamic Filtering, Metadata Routing, Query Expansion, Query Rewriting, Step-Back Retrieval, Citation & Source Grounding, Latency Optimization, Cold-Start Mitigation, Index Refresh Strategies, RAG Evaluation (Relevancy, Faithfulness, Context Recall, Answer Correctness using DeepEval, Ragas) Built GenAI workflows using **Copilot Studio** for automating internal business use cases.Used **Microsoft Fabric** for data ingestion, transformation, and analytics pipelines.

Key Projects

1. Agentic RAG for Financial Analysis(AWS)

Business Problem: Financial analysts spent excessive time gathering and validating data from multiple systems.
Value Delivered: Reduced analysis review time by 70% through autonomous agent collaboration.
Highlights:

- Architected multi-agent system using LangChain and LangFlow with 5 specialized agents — *Retrieval, Planning, Reasoning, Execution,* and *Delivery*.
- Deployed on Amazon EKS with horizontal autoscaling and standardized inter-agent schemas.
- Retrieval Agent: Implemented hybrid search (BM25 + vector) via Amazon OpenSearch with re-ranking and filter relaxation.
- Planning Agent: Applied few-shot task decomposition (avg. 2.6 steps) and deterministic prompts.
- Reasoning & Execution Agents: Used chain-of-thought prompting with Pandas and NumPy for safe computations.
- Delivery Agent: Generated audit-ready narratives with citations and integrated Amazon SES for automated reporting.

- Monitored KPIs via CloudWatch dashboards – Task Success: 93%, Error Recovery: 82%.

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2. PharmaRAG – Azure-Based Drug Information & Pharmacovigilance Chatbot

(Azure OpenAI | Azure AI Search | Azure Functions | Azure Document Intelligence | Azure Blob Storage | Streamlit | FastAPI | Azure DevOps)
Business Problem: Pharmacovigilance teams spent hours searching through unstructured drug data.
Value Delivered: Reduced lookup time by 70% and improved compliance with GxP, GVP, and HIPAA.
Highlights:

- Built Azure-native RAG pipeline for FDA SPL and RxNorm integration.
- Used Azure Document Intelligence for data extraction and AI Search for semantic retrieval.
- Integrated GPT-4o with FastAPI and Azure Functions for compliant real-time queries.
- Deployed Streamlit UI with Azure DevOps CI/CD automation.

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3. Healthcare MCP Server for Clinical Decision Support

Business Problem: Clinicians faced delays accessing structured healthcare data.
Value Delivered: Reduced drug–drug interaction lookup time by 70% with reliable, real-time data access.
Highlights:

- Developed Model Context Protocol (MCP) server linking AI models to regulated datasets (RxNorm, FAERS, ClinicalTrials.gov, FHIR).
- Built APIs for drug interaction, trial matching, and adverse-event summarization using Azure OpenAI.

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4. MCP “Pharmacy Ops” Assistant

Business Problem: Manual OTC evaluation slowed patient service.
Value Delivered: Reduced recommendation time from 5 minutes to 30 seconds while maintaining compliance.
Highlights:

- Automated OTC recommendation and copay estimation using LangChain + Azure OpenAI.

- Integrated OpenFDA APIs for contraindication validation and RxNorm for normalization.
- Exposed APIs via Azure API Management and deployed using Azure Container Apps.

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5. AI-Powered Drug Interaction Checker for Telemedicine

Business Problem: Telemedicine providers lacked automated interaction checks.
Value Delivered: Prevented 18% of potential adverse drug events during pilot phase.
Highlights:

- Developed real-time interaction service linked to FHIR-compliant EHRs and DrugBank APIs.
- Generated severity-ranked alerts with FDA citations for clinician validation.
- Deployed via Azure App Service with Application Insights and API Management.